INCH-POUND
MIL-DTL-24643/11D
22 August 2002
SUPERSEDING
MIL-C-24643/11C
14 March 1994

# DETAIL SPECIFICATION SHEET

## CABLE, ELECTRICAL, 600 VOLTS, TYPE LSSSF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (watertight)

| | |
|---|---|
| First - | Copper conductor - AWG size 300 MCM, class G stranding, uncoated (see note 1). |
| Second - | Separator may be used at manufacturer's option where required to provide free-stripping insulation. |
| Third - | Ethylene propylene rubber insulation. Minimum average wall thickness 0.075 inch. |
| Fourth - | Reinforcement at manufacturer's option. |
| Fifth - | Cross-linked polyolefin jacket bonded to the underlying insulation. Minimum average wall thickness 0.075 inch. |

NOTE: 1. The copper shall be hard drawn, having a minimum tensile strength of 67,200 lb/in$^2$, and minimum elongation of 0.85 percent in 60 inch lengths before stranding. The resistance specified is based on 102.7 percent of the values required for annealed copper.



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TABLE I. Details.

| Military part no. M24643/11 | Type and size | Conductor size AWG | Overall diameter | |
|---|---|---|---|---|
| | | | minimum (inches) | maximum (inches) |
| -01UN | LSSSF-7 | 300 MCM (class G) | 1.020 | 1.100 |

EXAMINATION AND TESTS:

Basic electrical: Requirements:

Conductor resistance - ohms/1000 feet at 25°C, maximum .....0.0388
Voltage withstand - volts, root mean square, minimum
Conductor to ground .....2500
Insulation resistance - conductor to conductor in water,
Megohms - 1000 feet, minimum .....100
Conductor continuity ..... No failure

Group A:

Visual and dimensional..... No failure

Group B:

Cross-linked proof test (percent, maximum) insulation .....50
Jacket (When tested at 200°C).....50
Tear - pounds per inch thickness, minimum (ASTM D 470) .....35
Cold bending, cable - at minus 30 ± 2°C, two turns around mandrel with the diameter eight times that of specimen ..... No damage

Physicals (unaged)

Insulation

Ethylene propylene rubber

Tensile strength - $lb/in^2$,minimum .....700
Elongation - percent, minimum .....250

Jacket (cable)

Tensile strength - $lb/in^2$, minimum .....1300
Elongation - percent, minimum .....160

Group C:

Physicals (aged)

Insulation

Ethylene propylene

Air oven

Tensile strength - percent of unaged, minimum .....75
Elongation - percent of unaged minimum .....75

Jacket (cable)

Air oven

Tensile strength - percent of unaged, minimum .....60
Elongation - percent of unaged minimum .....60



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Hot oil immersion
Tensile strength - percent of unaged, minimum ..... 50
Elongation - percent of unaged minimum ..... 50
Shrinkage ..... No failure
Permanence of printing (jacket) - cycles, minimum ..... 125
Heat distortion - percent of unaged, maximum ..... 30

Group D:
Flame propagation (cable) ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:
Aging and compatibility (cable) (125 ± 5°C) ..... No failure
Abrasion resistance (jacket) - scrapes, minimum ..... 75
Acid gas equivalent - percent, maximum
Jacket ..... 2
Fillers ..... 2
Insulation ..... 18
Halogen content - percent, maximum
Jacket ..... 0.2
Fillers ..... 0.2
Immersion (jacket)
Tensile strength - percent of unaged, minimum ..... 50
Elongation - percent of unaged, minimum ..... 50
Smoke index, maximum
Jacket ..... 25
Fillers ..... 45
Insulation ..... 45
Toxicity index, maximum
Jacket ..... 5
Fillers ..... 5
Insulation ..... 1.5
Durometer (jacket) - (type A) hardness, minimum ..... 80
Weathering (jacket) ..... No failure

UNIT ORDERING LENGTH:
Minimum 1045, maximum 1155 feet



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Custodians:
Army - MI
Navy - SH

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

Preparing Activity:
Navy - SH
(Project 6145-2308-009)



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